



Lab 02: Wireless Access Point

AP · Laptop0 · Laptop1 · Smartphone0 · Switch · PC1 · PC2

Lab #02

Topic: Wireless Networking

Tool: Cisco Packet Tracer

Cert: CompTIA Network+

1. Lab Objectives

- Configure Access Point with SSID 'Ahmed', WPA2-PSK, AES encryption.
- Connect Laptop0, Laptop1 and Smartphone0 wirelessly to the Access Point.
- Connect PC1 and PC2 via wired switch on the same network.
- Verify wireless clients can reach wired clients using ping.
- Relate to Network+ objective 2.4: Wireless technologies and configuration.

2. Network Topology

Left side: Access Point (AccessPoint0) with Laptop1, Laptop0 and Smartphone0 connected wirelessly (dashed lines). Right side: Switch with PC1 (Fa0/1) and PC2 (Fa1/1). The Access Point links to the Switch on Fa2/1 via wired Ethernet.

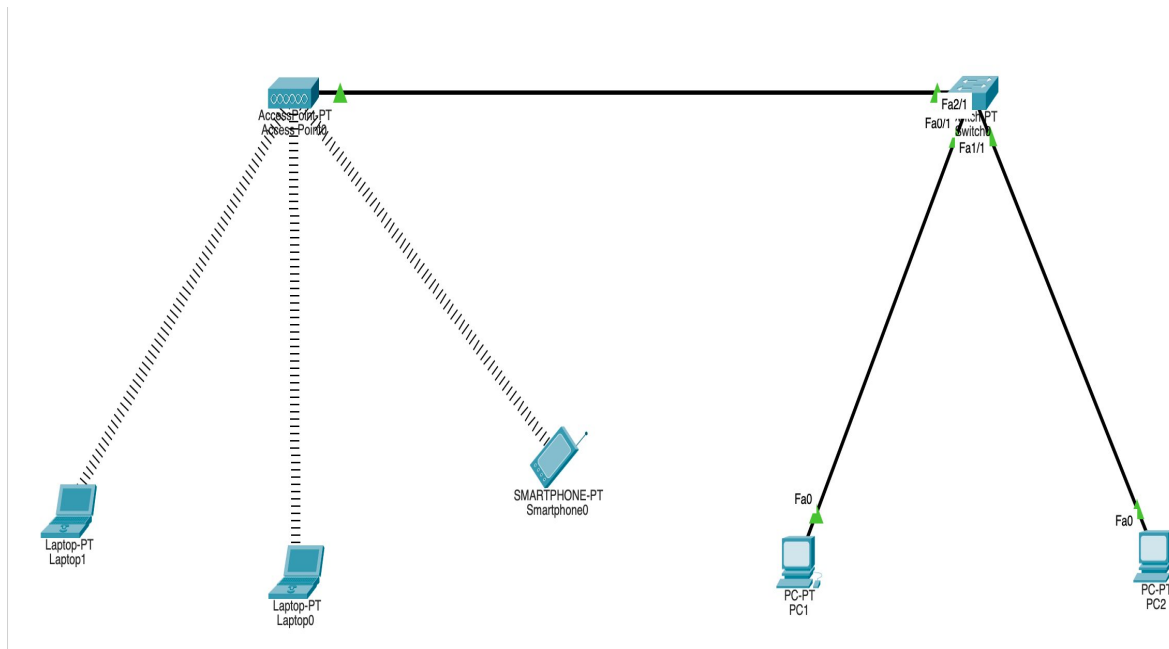


Figure 1 – Topology: Access Point, 3 wireless clients, Switch, PC1 & PC2

Device	Type	Connection	IP Range
AccessPoint0	AccessPoint-PT	Wired to Switch Fa2/1	192.168.10.0/24
Laptop1	Laptop-PT	Wireless – SSID Ahmed	DHCP



Laptop0	Laptop-PT	Wireless – SSID Ahmed	DHCP
Smartphone0	SMARTPHONE-PT	Wireless – SSID Ahmed	DHCP
PC1	PC-PT	Wired Switch Fa0/1	192.168.10.x
PC2	PC-PT	Wired Switch Fa1/1	192.168.10.x

3. Configuration

3.1 Access Point – Port 1 Settings

GLOBAL

Settings

INTERFACE

Port 0

Port 1

Port 1

Port Status ☒ On

SSID Ahmed

2.4 GHz Channel 6

Coverage Range (meters) 140.00

Authentication

☐ Disabled ☐ WEP ☒ WPA2-PSK

WEP Key

PSK Pass Phrase 12345678

User ID

Password

Encryption Type AES

Figure 2 – AP config: SSID Ahmed, Channel 6, WPA2-PSK, AES, PSK 12345678

SSID	Ahmed
Channel	6 (2.4 GHz)
Coverage	140 m
Authentication	WPA2-PSK
PSK Passphrase	12345678
Encryption	AES

3.2 Laptop Connection Steps

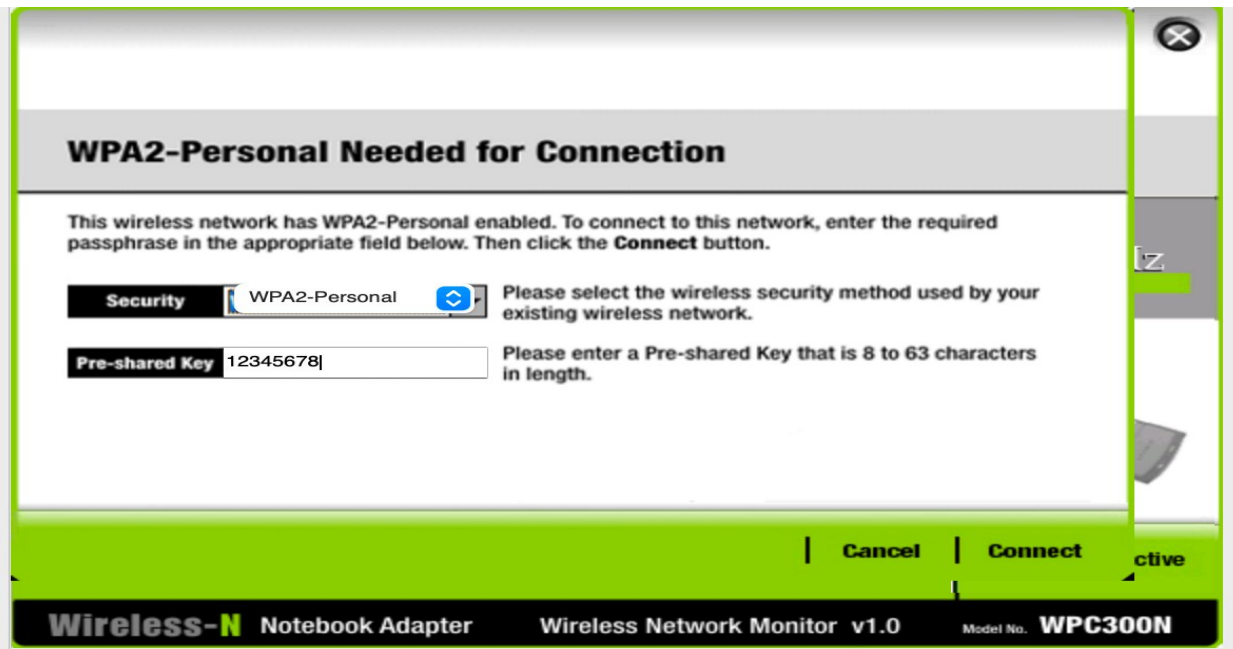


Figure 3 – WPA2-Personal prompt: Security WPA2-Personal, Pre-shared Key 12345678

1. Physical tab: remove Ethernet NIC, insert WPC300N wireless NIC
2. Desktop > PC Wireless > Connect tab > Refresh
3. Select SSID 'Ahmed' > Connect
4. Security: WPA2-Personal | Pre-shared Key: 12345678
5. Click Connect
6. Desktop > IP Configuration > set static or DHCP

4. Verification

```
Laptop0>ping 192.168.10.x ! to PC1 wired
Laptop1>ping Laptop0 ! wireless-to-wireless
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5. Key Takeaways

- WPA2-PSK with AES is the current Wi-Fi security standard — never use WEP.
- Channel 6 is one of three non-overlapping 2.4 GHz channels (1, 6, 11).
- An Access Point bridges wireless clients to the wired LAN at Layer 2.
- Coverage range (140 m) defines the AP signal radius in the simulation.
- Always install a wireless NIC in Packet Tracer laptops before connecting to an AP.